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ICS 2203 WEB APPLICATION DEVELOPMENT 1

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 2

Internet Architecture

Learning outcomes

Upon completing this topic, you should be able to:

1. Define the internet architecture
2. Describe how the internet works

2.1. Internet Addresses

Because the Internet is a global network of devices each device connected to the Internet must have a unique address. Internet addresses are in the form xxx.xxx.xxx.xxx where xxx must be a number from 0 - 255. This address is known as an IP address.



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(IP stands for Internet Protocol; more on this later.) The Figure below illustrates two computers connected to the Internet; your computer with IP address 1.2.3.4 and another computer with IP address 5.6.7.8. The Internet is represented as an abstract object in-between. (As this paper progresses, the Internet portion of Diagram 1 will be explained and redrawn several times as the details of the Internet are exposed).



Fig 1



2.2. Allocation of IP Addresses

If you connect to the Internet through an Internet Service Provider (ISP), you will be assigned a temporary IP address for the duration of your dial-in session. If you connect to the Internet from a local area network (LAN) your computer might have a permanent IP address or it might obtain a temporary one from a DHCP (Dynamic Host Configuration Protocol) server. In all these cases to the Internet, your device must have a unique IP address.

Example . Class Activity (in the lab) Try Out - The Ping Program

When using Windows Operating System or a Unix and have a connection to the Internet, there is a handy program to see



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if a computer on the Internet Live. It's called ping, If you are using Windows, start a command prompt window. Type ping www.yahoo.com. The ping program will send a 'ping' (actually an ICMP (Internet Control Message Protocol) echo request message) to the named computer. The pinged computer will respond with a reply. The ping program will count the time expired until the reply comes back (if it does). Also, if you enter a domain name (i.e. www.yahoo.com) instead of an IP address, ping will resolve the domain name and display the computer's IP address. More on domain names and address resolution later.



```
C:\Windows\system32\cmd.exe
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\brian>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\brian>
```

2.2.1. The TCP/IP Protocol Stack Suite for the Internet

The internet is a packet switching network. This means that the messages sent over the internet are usually broken down to smaller chunks that are transferrable over the internet. Let's say



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your IP address is 1.2.3.4 and you want to send a message to the computer 5.6.7.8. The message you want to send is "Hello computer 5.6.7.8!". The message must be translated from alphabetic text into electronic signals, transmitted over the Internet, then translated back into alphabetic text. This is accomplished through the use of a protocol stack. Every computer needs one to communicate on the Internet and it is usually built into the computer's operating system (i.e. Windows, Unix, etc.). The protocol stack used on the Internet is referred to as the TCP/IP protocol stack because of the two major communication protocols used.

Example . The table below illustrates TCP/IP stack .





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◀	▶
◀ Doc	Doc ▶
Back	Close



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If we were to follow the path that the message "Hello computer 5.6.7.8!" took from our computer to the computer with IP address 5.6.7.8, it would happen something like this:

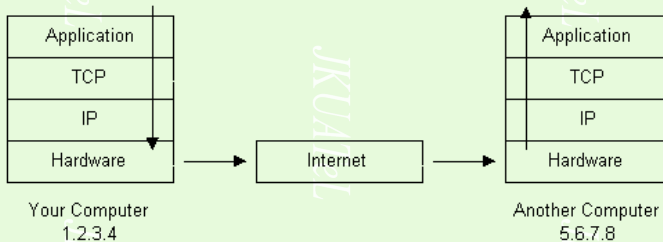


Diagram 2

1. The message would start at the top of the protocol stack on the application layer in your computer and work its way downward.
2. If the message to be sent is long, each stack layer that the



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message passes through may break the message up into smaller chunks of data known as packets.

3. The packets would go through the Application Layer and continue to the TCP layer. Each packet is assigned a port number, this is because we need to know which program on the destination computer needs to receive the message because it will be listening on a specific port.
4. After going through the TCP layer, the packets proceed to the IP layer. This is where each packet receives its destination address, 5.6.7.8.
5. Now that our message packets have a port number and an IP address, they are ready to be sent over the Internet. The hardware layer takes care of turning our packets containing the alphabetic text of our message into electronic



signals and transmitting them over the phone line.

6. On the other end of the phone line your ISP has a direct connection to the Internet. The ISP's router examines the destination address in each packet and determines where to send it. Often, the packet's next stop is another router.
7. Eventually, the packets reach computer 5.6.7.8. Here, the packets start at the bottom of the destination computer's TCP/IP stack and work upwards.
8. As the packets go upwards through the stack, all routing data that the sending computer's stack added (such as IP address and port number) is stripped from the packets.
9. When the data reaches the top of the stack, the packets have been re-assembled into their original form, "Hello



computer 5.6.7.8!”

2.2.2. The Network Infrastructure

Besides the knowledge of technologies in play to have a vast understanding of the internet we require some understanding on the actual layout of devices and medium involved from the backbone to the consumers of its services. Below is a representation of the layout of devices vis-à-vis other networks in cooperated.

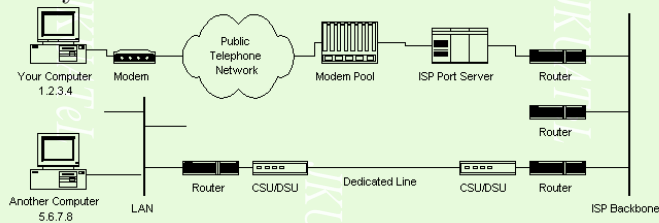


Diagram 3



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The ISP maintains a pool of modems for their dial-in customers. This is managed by some form of computer (usually a dedicated one) which controls data flow from the modem pool to a backbone or dedicated line router. This setup may be referred to as a port server, as it 'serves' access to the network. Billing and usage information is usually collected here as well. After your packets traverse the phone network and your ISP's local equipment, they are routed onto the ISP's backbone or a backbone the ISP buys bandwidth from. From here the packets will usually journey through several routers and over several backbones, dedicated lines, and other networks until they find their destination, the computer with address 5.6.7.8.



Components of Internet Network Architecture:

Satellite:

A major part of the internet network architecture is the satellite. Satellite plays a vital role in catching and distributing the signals over the network and the users use the internet network to search different types of information at any time.

Network Adapters:

There are different types of network adapters that are used to configure or setup the internet technology on your operating system. First install the network adapters in the system then install its software for the sake of its proper working or compatibility. Some common network adapters that are used for access of the internet are LAN cards or modems etc.



Routers:

As we know that this technology is also operates wirelessly so some components that are used to configure the internet network technology wireless router plays an important role and it is also the main part of the architecture. It is defined as the device that is used to transmit data from one place to another in the form of packets that are called as data packets is known as router. These data packets are also called data gram.

Access Points:

A special type of routing device that is used to transmit the data between wired and wireless networking device is called as AP. It is often connected with the help of wired devices such as Ethernet. It only transmits or transfers the data between wire-



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less internet technology and wired internet network technology by using infra structure mode of network. One access point can only support a small group of networks and works more efficiently. It is operated less than hundred feet. It is denoted by AP.

Clients:

Any kind of device such as personal computers, Note books, or any kind of mobile devices which are inter linked with wireless network area referred as a client of internet network architecture.

Bridges:

A special type of connectors which is used to establish connections between wired network devices such as Ethernet and different wireless networks such as wireless LAN. It is called as



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bridge. It acts as a point of control in internet network architecture. Two components are also some time play an important role in internet network architecture

1. Basic Service Set (BSS)
2. Extended Service Set (ESS)

Internet Infrastructure

The Internet backbone is made up of many large networks which interconnect with each other. These large networks are known as Network Service Providers or NSPs. Some of the large NSPs are UUNet, CerfNet, IBM, BBN Planet, SprintNet, PSINet, as well as others. These networks peer with each other to exchange packet traffic. Each NSP is required to connect to three Network Access Points or NAPs. At the NAPs, packet traffic may jump



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from one NSP's backbone to another NSP's backbone. NSPs also interconnect at Metropolitan Area Exchanges or MAEs. MAEs serve the same purpose as the NAPs but are privately owned. NAPs were the original Internet interconnect points. Both NAPs and MAEs are referred to as Internet Exchange Points or IXs. NSPs also sell bandwidth to smaller networks, such as ISPs and smaller bandwidth providers. Below is a picture showing this hierarchical infrastructure.

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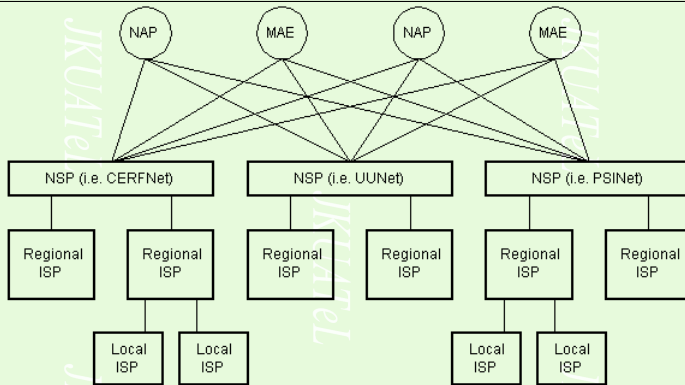


Diagram 4 This is not a true representation of an actual piece of the Internet. Diagram 4 is only meant to demonstrate how the NSPs could interconnect with each other and smaller ISPs. None of the physical network components are shown in Diagram 4 as they are in Diagram 3. This is because a single



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NSP's backbone infrastructure is a complex drawing by itself. Most NSPs publish maps of their network infrastructure on their web sites and can be found easily. To draw an actual map of the Internet would be nearly impossible due to it's size, complexity, and ever changing structure.

Example . Questioning individuals about drug use, sexual orientation, criminal activity, or other sensitive topics, is a difficult task. Individuals are not often willing to reveal such information for fear of social stigma, yet, for public-health and socioeconomic reasons, accurate information on such behaviors is often needed. In a telephone survey involving USC undergraduates, 700 individuals were asked to respond to the question: "Have you ever experimented with marijuana?" Results: Of these 700 individuals, 360 responded "yes," while 210 responded "no."



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There were 130 individuals which did not answer. Identify the population and the sample.

Solution: The population would be all the USC undergraduates while the sample is the 700 who were asked to respond ☐



Revision Questions

EXERCISE 1. 🖱 Using valid examples discuss the layout architecture of the internet.

EXERCISE 2. 🖱 Enumerate on the two tier and three tier client server architectures stating where each will be best suited. (2 marks)

Two tier architectures

Three tier architectures

EXERCISE 3. 🖱 The World Wide Web is based on the client server architecture (10 marks)

EXERCISE 4. 🖱 Describe how the www works (6 marks)



References and Additional Reading Materials

- George Beekman,(2008) Tomorrow's Technology and You, Complete, 9/E,Oregon State University Ben Beekman, ISBN-10: 0135045045 • ISBN-13: 9780135045046 ©2010 • Prentice Hall • Paper, 672 pp
- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X
- Craig Grannell(2007 The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5



Solutions to Exercises

Exercise 1.

i. The users PC, Mobile Phone, Black Berry equipped with Software (web browsers).

ii. The Users communication equipment i.e. modem, routers, Ethernet NIC, Servers etc – all equipment connecting user to the local loop

iii. Local Loop Carrier: - satellite, wireless, Fibre Optic Backbone, Telephone Cables. Common Companies that own run these include Telkom Kenya, Kenya Data Networks, Safaricom, Zain, Yu:- Controlled by CCK.

iv. ISP POP (Point of Presence):- concerned with authentication of user connections. You may need to be subscribed to



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an ISP to get services from them. Some ISP's include wananchi online, UUNet etc.

v. User Services - these are the services that most users use along with Internet Access. (These may be hosted within a large corporate LAN) (Webhosting is discussed under the online content section)-DNS (Domain Name Server) - BIND, DNS Resources Directory, Email Host -,Send mail ,Microsoft Exchange , Usenet Newsgroups (NNTP) - INN, Special services such as telnet, FTP Web Hosting, Data Back Up Services,

vi. ISP Backbone - The ISP backbone interconnects the ISP's POPs, AND interconnects the ISP to Other ISP's and online content. Research on providers of these in Kenya.

vii. Online Content: - These are the host sites that the user interacts with Web Server platforms - Netsite, Apache, Mi-



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rosoft, Yahoo's List of web servers. Hosting Firms- Many online resources are hosted at well-connection facilities.

viii. Origins of Online Content:- This is the original "real-world" sources for the online information. Existing electronic information is being connected from legacy systems. Traditional print resources are being scanned and converted into electronic format

Exercise 1



Exercise 2.

In two tier client/server architectures, the user interface is placed at user's desktop environment and the database management system services are usually in a server that is a more powerful machine that provides services to the many clients. Information processing is split between the user system interface environment and the database management server environment. The database management server supports for stored procedures and triggers. Software vendors provide tools to simplify development of applications for the two tier client/server architecture. The three tier architecture is introduced to overcome the drawbacks of the two tier architecture. In the three tier architecture, a middle-ware is used between the user system interface client environment and the database management server environment.



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These middle-ware are implemented in a variety of ways such as transaction processing monitors, message servers or application servers. The middle-ware perform the function of queuing, application execution and database staging. In addition the middle-ware adds scheduling and prioritization for work in progress. The three tier client/server architecture is used to improve performance for large number of users and also improves flexibility when compared to the two tier approach. The drawback of three tier architectures is that the development environment is more difficult to use than the development of two tier applications.

Exercise 2



Exercise 3.

this is the answer to the THIRD question

Exercise 3



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Exercise 4.

Every computer connected to the Internet has a unique address: a number whose format is defined by the Internet Protocol (IP), the standard that defines how messages are passed from one machine to another on the Net. An IP address is made up of four numbers, each less than 256, joined together by periods, such as 192.12.248.73 or 131.58.97.254. Each computer on the Internet also has a name bestowed upon it by its owner. Special computers, known as name servers, keep tables of machine names and their associated unique IP numerical addresses, and translate one into the other for us and for our machines. Domain names must be registered to be used.

Exercise 4